Curriculum Vitae

Personal Information

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Professional Summary

Honours Computer Science student at the University of the Western Cape with a strong interest in Cybersecurity, Artificial Intelligence, Cloud Computing, and Software Engineering. Experienced in designing IoT systems, developing full-stack applications, and building data-driven software solutions using Python, Java, JavaScript, Flask, MQTT, Linux, and Git.

Currently expanding technical expertise through industry internships and professional training, including the NITHECS June–July 2026 Internship Programme, the WeCnCode AI Cybersecurity Bootcamp, AI Engineering training through UWC Colabs, and AWS Cloud certification preparation. Passionate about solving real-world problems through secure, scalable technologies while contributing effectively within collaborative teams.

Experience

- Intern, NITHECS Internship Programme (June–July 2026)
  - Selected to participate in the June–July 2026 internship programme.
  - Collaborating on research and technical activities in computational science and emerging technologies.
  - Strengthening analytical, programming, and research skills through practical projects and mentorship.

- Tutor, UWC Faculty of Natural Sciences (Current)
  - Assisting 1st-year students in MAM115 (Calculus for Life Sciences)
  - Tutored first-year students in Differential and Integral Calculus with applications in Life Sciences and Pharmacy.
  - Assisted students with functions, limits, differentiation, integration, optimisation, curve sketching, and applied calculus problems.
  - Guided students through tutorial sessions, problem-solving techniques, and continuous assessment preparation.
  - Simplified complex mathematical concepts to improve student understanding and academic performance.

- Tutor, UWC Faculty of Natural Sciences (2025-2025)
  - Assisting 2nd-year students in MAT212 (Differential Equations & Numerical Methods)
    Supporting understanding of differential equations, numerical methods

(e.g., Newton's method, interpolation), and continuous assessment preparation

Education

- BSc (hons) Computer Science, University of the Western Cape(2026-current)

- BSc in Computer Science, University of the Western Cape (2021–2025)
  - Key subjects: Programming, Algorithms, Artificial Intelligence, Machine Learning, Software Engineering, Data Structures, Numerical Methods, Operating Systems, Databases, Computer Networks, Financial Mathematics, Mathematical Modelling

- National Senior Certificate, Basa Protea Glen Combined School (2019)
  - Key subjects: Mathematics, Physical Sciences, Life Sciences, Geography, English, Life Orientation, English, Afrikaans, Tourism

Professional Development

- **WeCnCode AI Cybersecurity Bootcamp** *(2026 – Current)*
  - Currently completing practical training in Cybersecurity.
  - Developing skills in secure software development, network security, vulnerability assessment, and security best practices.
- **AI Engineering Programme – UWC Colabs** *(2026 – Current)*
  - Currently completing professional training in AI Engineering.
  - Building practical experience in machine learning, generative AI, AI application development, and deployment.
- **AWS Certified Cloud Practitioner** *(In Progress)*
  - Working towards AWS Cloud certification.
  - Studying cloud architecture, security, networking, storage, and cloud deployment best practices.
- **Cybersecurity Professional Certifications**
  - Currently working towards internationally recognised cybersecurity certifications.

Projects

- **IoT Environmental Monitoring System**
  - Built a distributed system using sensors (temperature, soil moisture, gas) to collect environmental data
  - Used MQTT (Mosquitto) for real-time communication between devices
  - Set up and configured the system on a Linux-based device (Rock Pi)
  - Debugged connectivity and data transmission issues across the network
- **Real-Time IoT Dashboard**
  - Developed a dashboard using Flask to display live sensor data
  - Connected backend to MQTT topics for real-time updates
  - Designed simple UI pages to visualize environmental readings
  - Deployed with Nginx and resolved server and routing issues
- **Trading Tools Suite**
  - Built a pip calculator in Python for trading calculations and position sizing
  - Developed a JavaScript-based leaderboard to rank trader performance
  - Focused on clear data display and usability
  - Designed as a practical tool for quick decision-making
- **Cryptography Puzzle Engine**
  - Created a math-based cipher application using Python (Streamlit)
  - Implemented encryption techniques like Caesar and substitution ciphers
  - Used Firebase and JSON to store and manage puzzle data
  - Built an interactive interface for solving and testing cipher challenges
- **LaTeX Math Worksheet Generator**
  - Generated structured math worksheets (limits, exponents, logarithms) using LaTeX
  - Created large sets of problems with detailed solutions
  - Designed clean, professional layouts for academic use
  - Used for tutoring and helping students understand concepts

Leadership & Achievements

- Winner, Xylem Innovation Challenge #2 (2025) – CleanShield: Solar-powered IoT device for glacier black-carbon detection and community alerts via LoRa, AI/ML, SMS, and dashboards
- President, UWC Limpopo Society (2024-2025) – Led community initiatives for Limpopo students
- Treasurer, UWC Physics Society (2024- 2025) – Managed budgeting and financial oversight
- Graduate, Emerging Leaders Programme (UWC Student Leadership Academy, 2024) – Honed leadership, critical thinking, and problem-solving skills

Languages

Skills

Programming Languages

- Python
- Java
- C#
- JavaScript
- HTML
- CSS
- SQL

Cloud & DevOps

- AWS (Currently Studying)
- Linux
- Nginx
- Git
- GitHub

Cybersecurity

- Secure Software Development
- Cryptography
- Network Security Fundamentals
- Vulnerability Assessment (Training)
- Cybersecurity Best Practices

AI & Data Science

- Machine Learning
- Artificial Intelligence
- AI Engineering (Training)
- Data Analysis
- Mathematical Modelling

IoT

- MQTT
- Flask
- LoRa
- XBee
- Sensor Networks
- Embedded Systems

Soft Skills:
- Leadership & Team Management
- Strategic Planning & Organization
- Analytical Thinking & Problem-Solving
- Communication & Public Speaking
- Teamwork & Collaboration

Hobbies & Interests

- Robotics and IoT Projects – Enjoy building prototypes that integrate sensors and microcontrollers, exploring automation and smart solutions.
- Open-Source Contributions – Passionate about contributing to coding communities and improving collaborative projects.
- Reading AI/ML Research Papers – Keeps updated with the latest developments in artificial intelligence, deep learning, and data science.
- Community Leadership Initiatives – Actively engages in mentoring, organizing events, and empowering peers.

References

Dr. Grant E. Muller
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